

PlumeTrace

Autonomous Water Protection for the Pegnitz River

Why Pegnitz

Public Health

- Raw sewage spikes after rainfall
- Instant health risk for surfers, waders, and park visitors

Sewer Overflows (CSOs)

- 1,500 km municipal network overloaded by storms
- 1.5M - 5M+ EUR cost per retention basin

Knoblauchsland Agriculture

- Nitrate leaching into drinking water aquifers
- Strict EU limits and compliance pressure

Summer Heat and Drought

- Low flow concentrates pollutants
- Dissolved oxygen drops < 3 mg/L (fish kills)

Current Monitoring

Municipal Authorities

- Stadtentwässerung Nürnberg (SUN) and Wasserwirtschaftsamt (WWA)
- Temperature, pH, conductivity, turbidity
- Dissolved oxygen and nutrients (Nitrate, Phosphate)
- **Ammonium (NH4+):** primary raw sewage tracer

The Sampling Gap

- Automatic sensors: continuous, but confined to few fixed locations
- Manual grab sampling: often only 1x per month (24-48h lab delay)
- Convective storm overflows: last only 2 to 4 hours

The Result: **Plume arrives, causes harm, and clears undetected.**

Key Stakeholders

Water Authorities (WWA, SUN)

- Cannot locate pollution origin (fixed stations lack spatial tracking)
- Blind infrastructure spending (1.5M - 5M+ EUR per basin without plume data)
- Chronic pipe seepage undetected until structural failure
- Lab results arrive 24-48 hours too late for emergency action

Citizens and Surfers

- Zero real-time visibility into water safety
- Unknowingly enter water during toxic overflow events
- High risk of gastrointestinal and skin infections
- Delayed municipal reports provide no immediate guidance

System Architecture

Physical Platform

- TurtleBot3 surface robotics testbed
- ROS 2 Jazzy onboard computing
- LDS-02 LiDAR and overhead tracking
- Emulated chemical sensor payload

Simulation Mirror

- Gazebo Harmonic physics engine
- Bi-directional ROS-Gazebo bridge
- Ghost follower pose tracking
- Dynamic obstacle injection

3D Digital Twin

- CesiumJS with Bavarian LOD2 3D data
- Real-time WebSocket telemetry stream
- Geo-referenced pipe outfall health
- Interactive urban risk heatmaps

Key Architecture: **Physical platform, simulation mirror, and 3D geospatial twin synchronized in real time.**

Autonomous Closed Loop

4D Adaptive Retasking

- Dynamically recomputes waypoints in real time
- **Chemical Gradient:** homes upstream to leaking pipe
- **Rain Radar:** pre-positions robot before storm runoff
- **Data Freshness:** prioritizes unvisited river reaches

Safety and Resilience

- Dual obstacle cloud: reroute navigation vs collision stop
- Edge buffer: zero data loss during network dropouts
- Tri-fold sync: odometry, AMCL, and Gazebo alignment
- Dynamic Nav2 goal updates in under 35 seconds

Autonomy Paradigm: **Closed-loop intelligence replaces rigid pre-scripted patrol routes.**

Why Mobile Robotics?

Gradient Tracing

- Tracks concentration gradients upstream
- Locates exact leaking pipe instead of diluted mix

Asset Intelligence

- Links water chemistry to specific municipal outfall IDs
- Direct evidence for multi-million EUR basin upgrades

River Corridor Mapping

- LiDAR detects riverbank erosion and sediment build-up
- Supports EU renaturation compliance

Operational Economics

- One self-cleaning mobile robot replaces 50+ fixed probes
- Eliminates biofouling and flood destruction

The Core Value: **Actionable spatial intelligence where static networks are completely blind.**